# BACHELOR OF COMPUTER APPLICATIONS (BCA)

## SEMESTER VI - END SEMESTER EXAMINATION

May 2024

**Paper Code: BCA-605**

**Subject: Computer Graphics and Multimedia**

Time Allowed: 3 Hours | Maximum Marks: 75

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**General Instructions:**

1. Attempt all sections. Section A is compulsory; internal choice is given in Sections B and C.
2. Write answers in legible handwriting and number them exactly as in the question paper.
3. Draw neat, labelled diagrams wherever required.

### SECTION A (5 x 3 = 15 Marks)

*Answer all five questions. Each question carries 3 marks.*

1. Define raster graphics. [3]

2. What is a pixel? [3]

3. Explain the term resolution. [3]

4. What is a keyframe? [3]

5. Define anti-aliasing. [3]

### SECTION B (4 x 10 = 40 Marks)

*Answer any four questions. Each question carries 10 marks.*

1. Explain the DDA line drawing algorithm with an example. [10]

2. Explain 2D transformations: translation, rotation and scaling. [10]

3. Describe the RGB and CMYK colour models. [10]

4. Explain the types of animation with examples. [10]

### SECTION C (2 x 10 = 20 Marks)

*Answer any two questions. Each question carries 10 marks.*

1. Explain the Cohen-Sutherland line clipping algorithm with an example. [10]

2. Discuss lossy and lossless image compression techniques used in multimedia. [10]

*** *END OF PAPER* ***